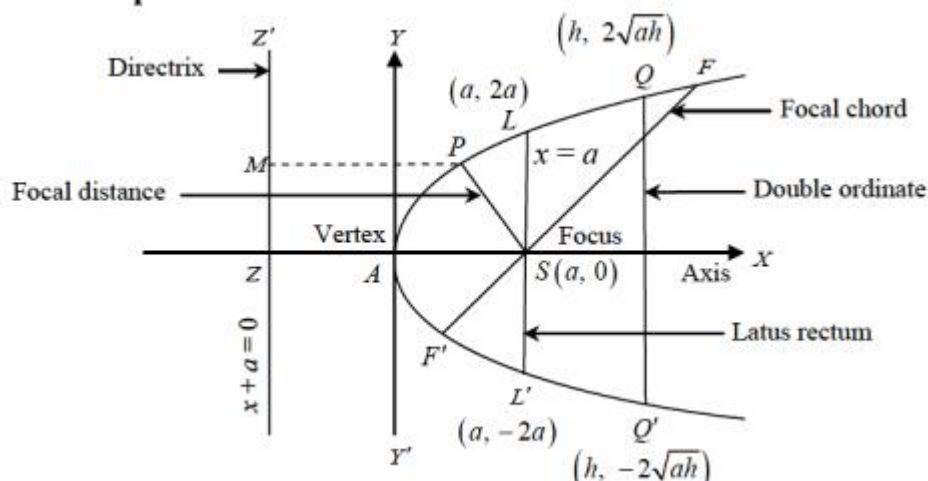


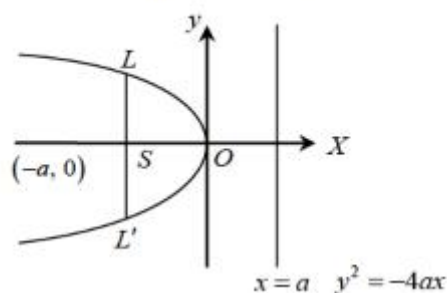
PARABOLA

EQUATION	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
FIGURE				
Vertex (A)	(0,0)	(0,0)	(0,0)	(0,0)
Focus (S)	(a,0)	(-a,0)	(0,a)	(0,-a)
Point of intersection of Axis and directrix (Z)	(-a,0)	(a,0)	(0,-a)	(0,a)
End point of latus rectum (L, L ₁)	(a, ±2a)	(-a, ±2a)	(±2a, a)	(±2a, -a)
Equation of axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
Equation of directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
Equation of tangent at Vertex	$x = 0$	$x = 0$	$y = 0$	$y = 0$
Equation of Latus rectum	$x = a$	$x = -a$	$y = a$	$y = -a$
Length of Latus rectum (LL ₁)	4a	4a	4a	4a
Distance from Focus to directrix (SZ)	2a	2a	2a	2a
SA = AZ	a	a	a	a

EQUATION	$(y - b)^2 = 4a(x - a)$	$(y - b)^2 = -4a(x - a)$	$(x - a)^2 = 4a(y - b)$	$(x - a)^2 = -4a(y - b)$
Vertex (A)	(a, b)	(a, b)	(a, b)	(a, b)
Focus (S)	(a + a, b)	(a - a, b)	(a, b + a)	(a, b - a)
Point of intersection of Axis and directrix (Z)	(a - a, b)	(a + a, b)	(a, b - a)	(a, b + a)
End point of latus rectum (L, L ₁)	(a + a, b ± 2a)	(a - a, b ± 2a)	(a ± 2a, b + a)	(a ± 2a, b - a)
Equation of axis	$y = b$	$y = b$	$x = a$	$x = a$
Equation of directrix	$x = a - a$	$x = a + a$	$y = b - a$	$y = b + a$
Equation of tangent at Vertex	$x = a$	$x = a$	$y = b$	$y = b$
Equation of Latus rectum	$x = a + a$	$x = a - a$	$y = b + a$	$y = b - a$
Length of Latus rectum (LL ₁)	4a	4a	4a	4a
Distance from Focus to directrix (SZ)	2a	2a	2a	2a
SA = AZ	a	a	a	a

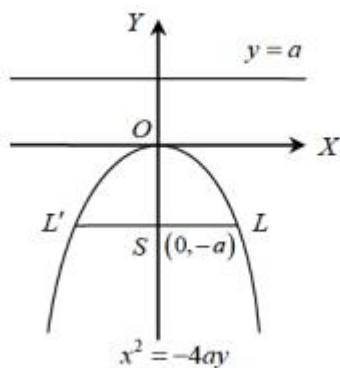
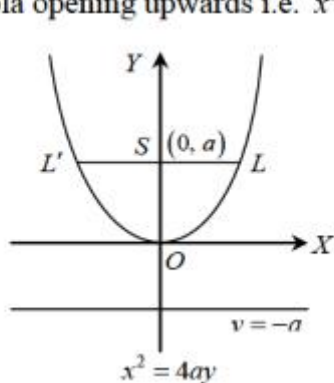


Parabola opening to left i.e. $y^2 = -4ax$



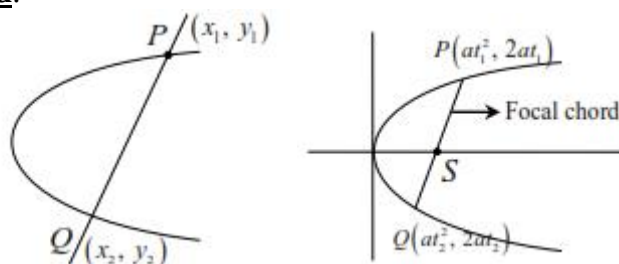
Parabola opening downwards i.e., $x^2 = -4ay$

Parabola opening upwards i.e. $x^2 = 4ay$



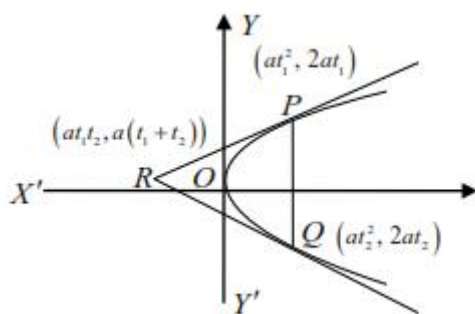
- The focal distance of the point $P(x_1, y_1)$ on the parabola $y^2 = 4ax$ is $SP = |x_1 + a|$.
- The focal distance of the point $P(x_1, y_1)$ on the parabola $x^2 = 4ay$ is $SP = |y_1 + a|$.
- A chord of the parabola perpendicular to its axis is called as **Double Ordinate** of the parabola.
- **Focal Chord:** A chord of the parabola which is passing through focus is called as focal chord.

- **Latus Rectum:** The focal chord of the parabola which is perpendicular to axis is called latus rectum.
- **Equation of Chord:**



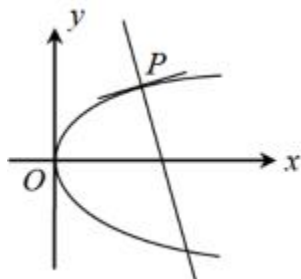
- (1) The equation of chord joining the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ on the parabola $y^2 = 4ax$ is $y(y_1 + y_2) = 4ax + y_1y_2$.
 - (2) The equation of chord joining the points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ is $y(t_1 + t_2) = 2(x + at_1t_2)$.
- **Condition for the chord to be a focal chord:** The chord joining $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ passes through the focus or it is called as focal chord if $t_1t_2 = -1$.
 - **Length of Focal Chord:** The length of a focal chord joining $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ is $PQ = a(t_1 - t_2)^2$.
 - **Equation of Tangents in Different forms:**
 - (1) **Point Form:** The equation of tangent to the parabola $y^2 = 4ax$ at $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$.
 - (2) **Parametric Form:** The equation of tangent to the parabola $y^2 = 4ax$ at the point $P(at^2, 2at)$ is $ty = x + at^2$.
 - (3) **Slope Form:** The equation of tangent to the parabola $y^2 = 4ax$ in terms of slope m is $y = mx + \frac{a}{m}$ or $m^2x - my + a = 0$ and the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.
 - (4) The condition of the line $y = mx + c$ may be a tangent to the parabola $y^2 = 4ax$ is $c = \frac{a}{m}$ and the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.
 - (5) For the parabola $y^2 = 4a(x + a)$, equation of tangent in slope form is $y = m(x + a) + \frac{a}{m}$.
 - (6) The condition of the line $y = mx + c$ may be a tangent to the parabola $x^2 = 4ay$ is $c = -am^2$ and the point of contact is $(2am, am^2)$.

- (7) The condition of the line $lx + my + n = 0$ may be a tangent to the parabola $y^2 = 4ax$ is $am^2 = l.n$ and the point of contact is $\left(\frac{n}{l}, -\frac{2am}{l}\right)$.
- (8) The condition of the line $lx + my + n = 0$ may be a tangent to the parabola $x^2 = 4ay$ is $al^2 = mn$ and the point of contact is $\left(-\frac{2al}{m}, \frac{n}{m}\right)$.
- (9) If m_1 and m_2 are the slopes of tangents from an external point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$ then they are the roots of $m^2 x_1 - my_1 + a = 0$ and hence $m_1 + m_2 = \frac{y_1}{x_1}$ and $m_1 m_2 = \frac{a}{x_1}$.
- (10) If θ be the angle between tangents drawn from $P(x_1, y_1)$ to the parabola $y^2 = 4ax$, then $\tan \theta = \frac{\sqrt{y_1^2 - 4ax_1}}{|x_1 + a|}$.
- (11) Locus of the point of intersection of perpendicular tangents drawn to a parabola is its directrix.
- (12) The angle between the tangents drawn from a point on the directrix to the parabola is 90° .
- (13) Tangents drawn at the ends of focal chord of a parabola are at right angles, they intersect on directrix.
- (14) The angle between the tangents drawn at the ends of latus rectum of the parabola $y^2 = 4ax$ is 90° and the point of intersection of these tangents is $(-a, 0)$.
- (15) The angle between the normals drawn at the ends of latus rectum of the parabola $y^2 = 4ax$ is 90° and the point of intersection of these normals is $(3a, 0)$.
- (16) Equation of common tangent to two parabola $y^2 = 4ax$ and $x^2 = 4by$ is $a^{1/3}x + b^{1/3}y + (ab)^{2/3} = 0$.
- (17) The tangents and normals at the ends of latus rectum of parabola $y^2 = 4ax$ form a square whose side is $2\sqrt{2}a$ and its area is $8a^2$.
- (18) Slope of the tangent at $(at^2, 2at)$ on the parabola $y^2 = 4ax$ is $\frac{1}{t}$.



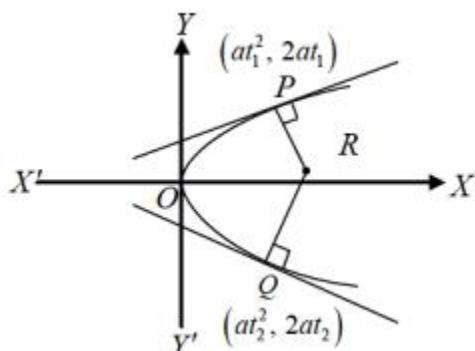
- **Points of intersection of Tangents:** The point of intersection of tangents drawn at two different points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ is $R(at_1t_2, a(t_1 + t_2))$.

- **Equation of Tangents in Different forms:**

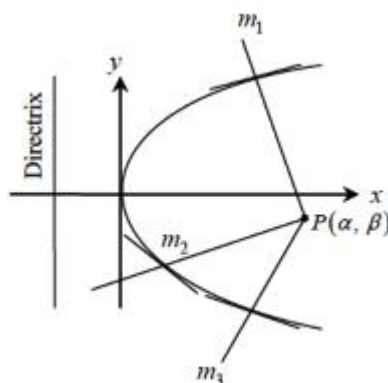


- (1) **Point Form:** The equation of tangent to the parabola $y^2 = 4ax$ at $P(x_1, y_1)$ is $y - y_1 = -\frac{y_1}{2a}(x - x_1)$.
- (2) **Parametric Form:** The equation of tangent to the parabola $y^2 = 4ax$ at the point $P(at^2, 2at)$ is $y + tx = 2at + at^3$.
- (3) **Slope Form:** The equation of tangent to the parabola $y^2 = 4ax$ in terms of slope m is $y = mx - 2am - am^3$ and the point of contact is $(am^2, -2am)$.
- (4) The condition that the line $y = mx + c$ to be normal to the parabola $y^2 = 4ax$ is $c = -2am - am^3$.
- (5) The condition that the line $lx + my + n = 0$ to be normal to the parabola $y^2 = 4ax$ is $al^3 + 2alm^2 + m^2n = 0$.
- (6) If the normal at $(at_1^2, 2at_1)$ on $y^2 = 4ax$ meets it again at $(at_2^2, 2at_2)$ then $t_2 = -t_1 - \frac{2}{t_1}$.
- (7) If the normals at $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ meet on parabola again at $(at_3^2, 2at_3)$ then $t_1t_2 = 2$ and $t_1 + t_2 + t_3 = 0$.
- (8) Slope of the normal at $(at^2, 2at)$ on the parabola $y^2 = 4ax$ is $-t$.
- (9) Equation of pair of tangents drawn from an external point (x_1, y_1) to parabola $y^2 = 4ax$ is $(y^2 - 4ax)(y_1^2 - 4ax_1) = (yy_1 - 2a(x + x_1))^2$.
- (10) If $(at_1^2, 2at_1)$, $(at_3^2, 2at_3)$ and $(at_2^2, 2at_2)$ are the feet of the normals drawn from a point (x_1, y_1) to the parabola $y^2 = 4ax$ then they are the roots of $at^3 + (2a - x_1)t - y_1 = 0$ and hence $t_1 + t_2 + t_3 = 0$, $t_1t_2 + t_3t_1 + t_2t_3 = \frac{2a - x_1}{a}$, $t_1t_2t_3 = \frac{y_1}{a}$.

- **Points of intersection of Tangents:** The point of intersection of normals drawn at two different points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola $y^2 = 4ax$ is $R(2a + a(t_1^2 + t_2^2 + t_1t_2), -at_1t_2(t_1 + t_2))$.



- **Co-Normal Points:** Any three points on a parabola normals at which pass through a common point are called a co-normal points. If three normals through a point $P(a, b)$, then their slopes are the roots of the cubic equation $b = ma - 2am - am^3$.



- (1) The sum of the slopes of the normals at co-normal point is zero i.e. $m_1 + m_2 + m_3 = 0$.
- (2) The sum of the co-ordinates of the co-normal point is zero i.e. $-2am_1 - 2am_2 - 2am_3 = -2a(m_1 + m_2 + m_3) = 0$.
- (3) The centroid of the triangle formed by the co-normal points lies on the axis of the parabola. The vertices of the triangle is $(am_1^2, -2am_1)$, $(am_2^2, -2am_2)$ and $(am_3^2, -2am_3)$. So,

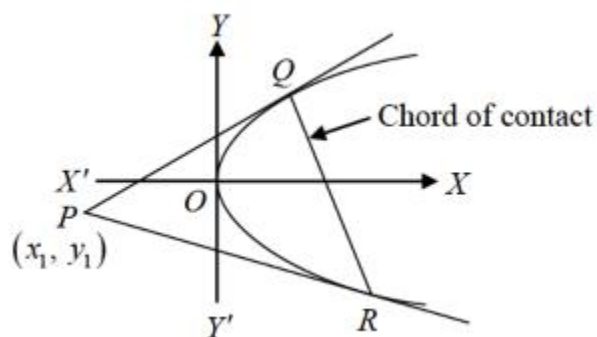
$$\text{centroid of the triangle is } \left(\frac{am_1^2 + am_2^2 + am_3^2}{3}, \frac{-2am_1 - 2am_2 - 2am_3}{3} \right)$$

$$\Rightarrow \text{centroid of the triangle is } \left(\frac{a(m_1^2 + m_2^2 + m_3^2)}{3}, \frac{-2a(m_1 + m_2 + m_3)}{3} \right)$$

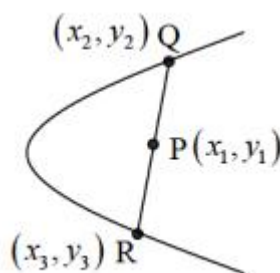
$$\Rightarrow \text{centroid of the triangle is } \left(\frac{a(m_1^2 + m_2^2 + m_3^2)}{3}, 0 \right)$$

- (4) If three normals drawn to any parabola $y^2 = 4ax$ from a given $(h, 0)$ be real, then $h > 2a$.

- **Position of a Point with respect to a parabola:** Let $P(x_1, y_1)$ be a point and $y^2 = 4ax$ be a parabola, then
 - (1) $P(x_1, y_1)$ lies on parabola $y^2 = 4ax$ if $y_1^2 - 4ax_1 = 0$.
 - (2) $P(x_1, y_1)$ lies inside parabola $y^2 = 4ax$ if $y_1^2 - 4ax_1 < 0$.
 - (3) $P(x_1, y_1)$ lies outside parabola $y^2 = 4ax$ if $y_1^2 - 4ax_1 > 0$.
- **Equation of Chord of Contact of Tangent to a parabola:** Let PQ and PR be tangents to the parabola $y^2 = 4ax$ drawn from any external point $P(x_1, y_1)$, then QR is called as the chord of contact of the parabola $y^2 = 4ax$. The chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the parabola $y^2 = 4ax$ is $yy_1 = 2a(x + x_1)$



- **Equation of the chord of the parabola which is bisected at a given point:** The equation of the parabola $y^2 = 4ax$ bisected at the point $P(x_1, y_1)$ is given by $yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$.



- (1) Equation of chord joining $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ on parabola $y^2 = 4ax$ is $y(t_1 + t_2) = 2x + 2at_1t_2$.
- (2) If $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ are the end points of focal chord of parabola $y^2 = 4ax$, then $t_1t_2 = -1$.

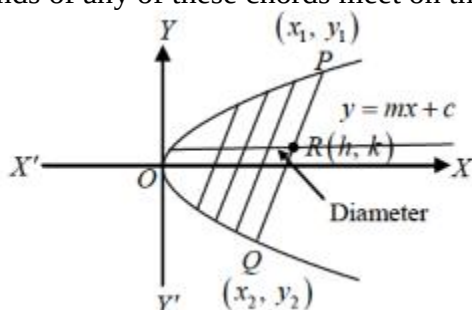
- (3) If $(at^2, 2at)$ is one end point of focal chord of the parabola $y^2 = 4ax$, then its other end is $\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$.
- (4) The tangent at end of focal chord of parabola is parallel to normal at the other end.
- (5) If the normal chord at $(at^2, 2at)$ on parabola $y^2 = 4ax$ subtend a right angle at the vertex then $t^2 = 2$.
- (6) If the normal chord at $(at^2, 2at)$ on parabola $y^2 = 4ax$ subtend a right angle at the focus then $t^2 = 4$.
- (7) If $P(x_1, y_1)$ and $Q(x_2, y_2)$ are the ends of chords of a curve then its length $PQ = |x_1 - x_2|\sqrt{1+m^2}$ where m is the slope of PQ.
- (8) If $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ are the ends of chord of parabola $y^2 = 4ax$ then its length $= |t_1 - t_2|\sqrt{(t_1 + t_2)^2 + 4}$.
- (9) The length of focal chord drawn at a point $(at^2, 2at)$ on the parabola $y^2 = 4ax$ is $a\left(t + \frac{1}{t}\right)^2$.
- (10) If a focal chord of parabola $y^2 = 4ax$ makes an angle θ with its axis then its length $= 4a \cdot \operatorname{cosec}^2 \theta$.
- (11) The length of normal chord at $(at^2, 2at)$ on $y^2 = 4ax$ is $\frac{4a(1+t^2)^{3/2}}{t^2}$.
- (12) Length of chord of contact of $P(x_1, y_1)$ with respect to $y^2 = 4ax$ is $\frac{1}{a}\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}$.
- (13) Length of chord of $y^2 = 4ax$ having $P(x_1, y_1)$ as its mid point is $\frac{1}{a}\sqrt{(-y_1^2 + 4ax_1)(y_1^2 + 4a^2)}$.
- (14) Area of triangle formed by two tangents from $P(x_1, y_1)$ and its chord of contact is $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$.
- (15) If PSQ is a focal chord of a conic then $\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}$ when l is length of semi latus rectum of a conic. OR The semi latus rectum is harmonic mean between segments of focal chord of the parabola.
- (16) The focal distance of a point $P(at^2, 2at)$ on the parabola $y^2 = 4ax$ is $|at^2 + a|$.
- (17) The length of the normal chord of the parabola is least when it subtends a right angle at vertex.

- **Diameters of a Parabola:** The locus of the middle points of a system of parallel chords is called a diameter of a parabola. The diameter is a straight line parallel to the axis of the parabola.

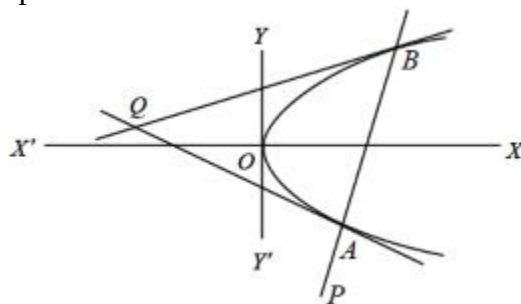
The equation of the diameter bisecting chords of the parabola $y^2 = 4ax$ of slopes m is

$$y = \frac{2a}{m}.$$

Tangents drawn at the ends of any of these chords meet on the diameter of these chords.



- **Poles and Polar of the Parabola:** Let P be a given point. Let a line through P intersects the parabola at two points A and B. Let the tangents at A and B intersect at Q. The locus of point Q is a straight line called as the polar of point P with respect to the parabola and the point P is called the pole of the polar.



(1) The polar of a point $P(x_1, y_1)$ with respect to the parabola $y^2 = 4ax$ is $yy_1 - 2a(x + x_1) = 0$.

(2) Pole of line $lx + my + n = 0$ with respect to parabola $y^2 = 4ax$ is $\left(\frac{n}{l}, -\frac{2am}{l}\right)$.

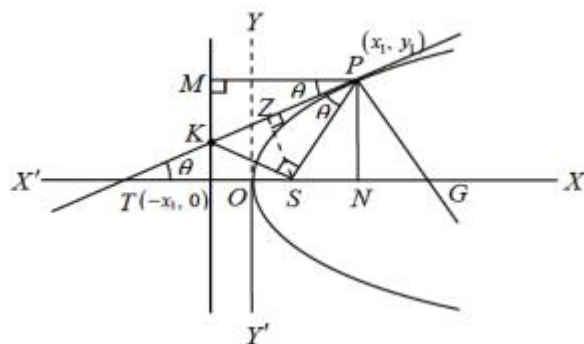
(3) Pole of line $lx + my + n = 0$ with respect to parabola $x^2 = 4ay$ is $\left(-\frac{2al}{m}, \frac{a}{m}\right)$.

- **Conjugate Points:** If the polar $P(x_1, y_1)$ passes through $Q(x_2, y_2)$, then the polar of $Q(x_2, y_2)$ passes through $P(x_1, y_1)$ and such points are called as **conjugate points**.

(1) The condition for two points (x_1, y_1) and (x_2, y_2) to be conjugate to parabola $y^2 = 4ax$ is $y_1 y_2 = 2a(x_1 + x_2)$.

- **Conjugate Lines:** If the pole of a line $l_1x + m_1y + n_1 = 0$ lies on the another line $l_2x + m_2y + n_2 = 0$, then the pole of line $l_2x + m_2y + n_2 = 0$ lies on line $l_1x + m_1y + n_1 = 0$. Such lines are called as **conjugate lines**.
 - (1) The condition that two lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate with respect to parabola $y^2 = 4ax$ is $l_1n_2 + l_2n_1 = 2am_1m_2$.
 - (2) The condition that two lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate with respect to parabola $x^2 = 4ay$ is $m_1n_2 + m_2n_1 = 2al_1l_2$.
- **Properties of Pole and Polar:**
 - (1) Polar of the focus is the directrix.
 - (2) Any tangent is the polar of its point of contact.
 - (3) The point of intersection of the polars of two points Q and R is the pole of QR.
 - (4) The pole of any focal chord lies on its directrix.
- **Properties of Parabola:**
 - (1) The equation $x = at^2$, $y = 2at$ are called as parametric form of parabola $y^2 = 4ax$. The point $(at^2, 2at)$ lies on the parabola $y^2 = 4ax$ for any value of t .
 - (2) The equation $y = at^2$, $x = 2at$ are called as parametric form of parabola $x^2 = 4ay$. The point $(2at, at^2)$ lies on the parabola $x^2 = 4ay$ for any value of t .
 - (3) If $P(at^2, 2at)$ is one of the end point of double ordinate of parabola $y^2 = 4ax$ then its length is $4at$.
 - (4) Area of triangle inscribed in parabola $y^2 = 4ax$ is $\frac{1}{8a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$ where y_1, y_2, y_3 are the ordinates of angular points.
 - (5) Area of the triangle formed by the tangents at three points whose ordinates are y_1, y_2, y_3 on $y^2 = 4ax$ is $\frac{1}{16a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$.
 - (6) The area of the triangle formed by any three points on the parabola is twice the area of the triangle formed by the tangents at these points.
 - (7) If SP and SQ are the distances of two points P and Q on parabola $y^2 = 4ax$ from focus S, T is a point of intersection of tangents at P and Q then $ST^2 = SP \cdot SQ$.
 - (8) The locus of the point of intersection of perpendicular tangents to the parabola is its directrix.
 - (9) The orthocenter of the triangle formed by three tangents of a parabola lies on directrix.
 - (10) The tangents at the extremities of any focal chord intersect at right angles on the directrix.
 - (11) The circle described on any focal chord of a parabola as diameters touches the directrix.
 - (12) If the normals t_1 and t_2 meet the parabola then $t_1t_2 = 2$.

- (13) A diameter of the parabola is parallel to its axis and the tangent at the point where it meets the parabola is parallel to the system of chords bisected by the diameter and tangent at the ends of any of the parallel chords of this diameter meet on the diameter.
- (14) The harmonic mean between the focal radii of any focal chord of a parabola is equal to semi latus rectum.
- (15) If the tangent and normal at any point P on the parabola meet the axis of parabola at T and G then:



- (a) $ST = SG = SP$, where S is the focus.
- (b) $\angle PSK = \frac{\rho}{2}$, where K is the point where the tangent at P meets the directrix.
- (c) The tangent at P is equally inclined to the axis of the parabola and the focal distance of P.
- (d) Length of sub-normal is always of constant length and is equal to the semi latus rectum of the parabola i.e. $2a$.
- (16) If we draw a circle taking any focal radius as diameter will touch the tangent at the vertex.
- (17) The foot of perpendicular from the focus on any tangent to the parabola is the point where the tangent meets the tangent at vertex. i.e. Z lies on y-axis and $SZ = \sqrt{OS.SP}$.
